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INSTRUCTIONS

1. Submission is only via Brightspace. Deadlines are strict.
2. You must submit a PDF typeset in (La)TEX using **this** template. Handwritten solutions will not be accepted.
3. Seeking solutions from the internet, from any external resource, or from any other person is prohibited.
4. Please note that the course lecturer reserves the right to ask the student submitting the assignment to explain the answers to any or all questions. If the student is unable to provide a satisfactory answer then that question may receive partial/no credit.
5. Of course, university policies on plagiarism always apply. In particular, any suspected plagiarism will be reported to the Board of Examiners.

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1. **Deadlock vs. Safe States** **(6 points)**

   Can a system be in a state that is *neither* deadlocked *nor* safe? If so, give a concrete example. If not, prove that every state is either deadlocked or safe.

   Solution:

2. **Multi-Unit Resource Deadlock** **(6 points)**

   Is it possible for a resource deadlock to involve multiple units of one resource type and a single unit of another? If so, give a concrete example showing the processes, resources, current allocations, and outstanding requests. If not, explain why.

   Solution:

3. **Full Virtualisation vs. Paravirtualisation** **(6 points)**

   (a) (3 points) Explain the difference between full virtualisation and paravirtualisation. Address how each handles privileged instructions, whether the guest OS needs modification, and give a concrete example.

   Solution:

   (b) (3 points) Which of the two is harder to implement? Justify with at least two technical challenges.

   Solution:

4. **Type 1 vs. Type 2 Hypervisors** **(6 points)**

   Type 1 hypervisors can do everything Type 2 hypervisors can, and are generally more efficient. So why do Type 2 hypervisors exist at all? Give at least three concrete reasons why a user or organisation might prefer a Type 2 hypervisor despite its lower efficiency.

Solution:

5. **Key Establishment with a Trusted Third Party** **(6 points)**

Suppose Alice and Bob have never met, but there exists a trusted third party, Trent, who shares a secret key K_{AT} with Alice and a different secret key K_{BT} with Bob. Describe a protocol by which Alice and Bob can establish a new shared session key K_S using only symmetric cryptography and the help of Trent.

Your protocol should:

1. Number each message.
2. State the sender, receiver, and contents.
3. Explain why an eavesdropper cannot recover K_S.

Solution: